

Time, speed & distance

COMPLETE LEARNING GUIDE

What You'll Learn

Time-speed-distance (TSD) problems test the same core relationship — distance = speed × time — in progressively disguised forms: relative speed between moving bodies, average speed over unequal time or distance segments, trains crossing objects of their own length, boats moving with or against a current, and circular-track races. Nearly every failure on this topic traces back to one of two errors: forgetting to convert units consistently (km/h vs m/s) or applying a simple average where a weighted or harmonic average is actually required. This sub-module builds fluency with the core formula and its unit conversions, then works through every major disguised variant — relative speed, average speed, trains, boats and streams, and races — with a fully worked numerical example for each, so that recognizing which variant a word problem represents becomes automatic rather than something to puzzle out under time pressure.

1. Core Formula and Unit Conversion

Distance = Speed × Time underlies every problem in this topic, but most errors come from inconsistent units. Converting km/h to m/s multiplies by $5/18$; converting m/s to km/h multiplies by $18/5$. Locking this conversion factor in as a reflex avoids a huge share of careless errors.

- Convert 72 km/h to m/s: $72 \times 5/18 = 20$ m/s. Convert 20 m/s back to km/h: $20 \times 18/5 = 72$ km/h. '

2. Relative Speed: Same Direction vs Opposite Direction

When two bodies move toward each other, their speeds add (relative speed = $s_1 + s_2$) and the time to meet = distance/($s_1 + s_2$). When moving in the same direction, relative speed is the difference ($s_1 - s_2$), used for 'catching up' problems.

- Two trains 120 km apart move toward each other at 60 km/h and 40 km/h. Time to meet = $120/(60+40) = 1.2$ hours = 72 minutes.
- A is 20 km ahead of B; A moves at 60 km/h and B chases at 40 km/h in the same direction. Time for B to catch up = $20/(60 - 40) = 1$ hour.

3. Average Speed Is a Harmonic Mean, Not a Simple Average

When equal DISTANCES are covered at two different speeds, average speed = $2 \times s_1 \times s_2 / (s_1 + s_2)$ — the harmonic mean, which is always less than the simple average of the two speeds. (If equal TIME is spent at each speed instead, the simple average is correct — the distinction matters.)

- A travels half the distance at 60 km/h and the other half at 40 km/h. Average speed = $2 \times 60 \times 40 / (60 + 40) = 4800 / 100 = 48$ km/h — NOT the simple average of 50 km/h.

4. Trains Crossing Poles, Platforms, and Each Other

A train crossing a stationary point (pole/man) travels its own length; crossing a platform, it travels its own length PLUS the platform's length. When two trains cross each other, use relative speed (sum if opposite directions, difference if same direction) and total length = sum of both train lengths.

- A 150 m train crosses a pole in 10 seconds: speed = $150 / 10 = 15$ m/s = 54 km/h.
- The same train (150 m, 15 m/s) crosses a 250 m platform: total distance = 400 m, time = $400 / 15 \approx 26.7$ seconds.
- A 100 m train and a 150 m train run toward each other at 40 km/h and 50 km/h. Relative speed = 90 km/h = 25 m/s; total length = 250 m; time to cross = $250 / 25 = 10$ seconds.

5. Boats and Streams

If boat speed in still water is b and stream speed is s , downstream speed = $b + s$ and upstream speed = $b - s$. Round-trip problems require calculating each leg's time separately (distance/downstream speed, then distance/upstream speed) and summing — they cannot be averaged directly.

- Boat speed in still water = 10 km/h, stream speed = 2 km/h. Downstream = 12 km/h, upstream = 8 km/h. For a 24 km trip each way: downstream time = $24 / 12 = 2$ h, upstream time = $24 / 8 = 3$ h. Total round trip = 5 hours.

6. Races and Head Starts

'A gives B a start of x meters in a race of d meters' means when A finishes d meters, B has covered $(d - x)$ meters in the same time — giving the speed ratio of A:B directly as $d : (d - x)$.

- A can give B a start of 20 m in a 100 m race (both finish together). Speed ratio A:B = $100 : 80 = 5 : 4$.

7. Circular Track Meeting Problems

On a circular track of length L , two runners moving in the same direction meet again after time = $L / (s_1 - s_2)$; moving in opposite directions, they meet after time = $L / (s_1 + s_2)$. This is structurally identical to the linear relative-speed rule, just wrapped around a loop.

- Two runners on a 300 m circular track run in the same direction at 5 m/s and 3 m/s. They meet again after $300/(5 - 3) = 150$ seconds.

Common Pitfalls

- Forgetting to convert km/h to m/s (or vice versa) before combining with a distance given in a different unit.
- Averaging two speeds arithmetically when equal DISTANCE (not equal time) was covered at each speed — this needs the harmonic mean, $2s_1s_2/(s_1+s_2)$.
- Forgetting to add the platform's length to the train's own length when computing platform-crossing time.
- Using the sum of speeds for same-direction 'catching up' problems instead of the difference.
- In boats and streams, averaging upstream and downstream speed directly instead of calculating each leg's time separately when distances are equal but speeds differ.
- Misreading a race 'start' or 'beats by' statement and inverting the resulting speed ratio.

Key Takeaways

- Lock in the conversion: $\text{km/h} \times 5/18 = \text{m/s}$, and $\text{m/s} \times 18/5 = \text{km/h}$.
- Opposite-direction relative speed adds; same-direction relative speed subtracts.
- Average speed for EQUAL DISTANCES is the harmonic mean $2s_1s_2/(s_1+s_2)$, always less than the simple average.
- Train crossing a platform: distance = train length + platform length; crossing a pole: distance = train length only.
- Boats and streams: downstream = $b+s$, upstream = $b-s$; compute each leg's time separately for round trips.
- A 'start of x in a race of d' gives speed ratio $d:(d-x)$.

Learn More

IndiaBix — Time and Distance Aptitude Questions

<https://www.indiabix.com/aptitude/time-and-distance/>

IndiaBix — Boats and Streams

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